# Christian Carrico

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## Experience

**The Aerospace Corporation** — Colorado Springs, CO
*Modeling & Simulations Systems Architect* — *Mar 2026 – Present*

**SPOT Award**

**U.S. Army DEVCOM Aviation & Missile Center** — Colorado Springs, CO
*Technical Lead* — *Mar 2025 – Mar 2026*

- Led engineering teams of government, FFRDC, and contractor personnel to deliver AI/ML research as operational software capabilities for the Warfighter
- Developed and presented technical roadmaps to senior leadership, communicating milestones, risks, and challenges to guide strategic decisions
- Briefed potential stakeholders on project capabilities to drive engagement and buy-in from COCOMs and other DoD organizations

**Letter of Recognition** from **Software, Simulations, and Systems Integration Director, SES**

**Certificate of Achievement** from **Aviation and Missile Center Director, SES**

**U.S. Army DEVCOM Aviation & Missile Center** — Colorado Springs, CO
*Software Engineer* — *Apr 2018 – Mar 2025*

- Employed Amazon Web Services to deploy critical infrastructure such as CI/CD pipelines and collaboration tools to support various government efforts
- Leveraged technologies such as the Atlassian tool suite and GitLab to capture customer requirements, document design decisions, and accelerate the development process
- Partnered with small businesses on SBIR/STTR contracts, overseeing progress and deliverables while providing technical guidance to maximize government value
- Utilized SysML and Cameo to develop system models, supporting early-stage architecture design, analysis, and CONOPs
- Mentored student interns and new employees by delegating meaningful tasks that foster skill development and professional growth
- Evaluated research proposals for technical and commercial merit while ensuring alignment with Agency priorities

## Projects

**AFSIM Reinforcement Learning Environment**

- Designed and delivered solutions for a Python Reinforcement Learning environment that integrates AFSIM and graph based observations to optimize blue force outcomes
- Developed code to procedurally generate sufficiently random AFSIM scenarios, providing a reinforcement learning agent with enough experiences to enhance generalization across the entire problem space
- Built Regex- and Jinja2-powered scripts to parse hundreds of AFSIM model files into a Python interface, enabling better scenario generation and a richer feature space
- Optimized Docker and Docker Compose configurations, reducing environment development image size by 60%, dramatically reducing build times and accelerating deployments to a classified network

**Modeling & Simulation Framework Trade Study**

- Directed research and assessment of multiple M&S frameworks, coordinating tasks with government and contractor personnel
- Developed objective evaluation criteria and formalized use cases to determine the most suitable framework
- Compiled findings into a comprehensive report and presented key insights to leadership for decision-making

## Education

**University of Colorado Colorado Springs** Colorado Springs, CO
*M.E. in Space Operations* *June 2026 – Present*

**Colorado State University** Fort Collins, CO
*B.S. in Computer Engineering* *Aug 2012 – Dec 2017*
*B.S. in Mathematics, Concentration in Information Security* *Aug 2012 – Dec 2017*